PREECLAMPSIA SYNDROME

1. OVERVIEW

Definition and Importance

- **Preeclampsia is a pregnancy-specific multisystem disorder characterized by hypertension and systemic endothelial dysfunction.**
- **Hypertensive disorders complicate up to 10% of pregnancies and are a major cause of maternal morbidity and mortality.**
- **Preeclampsia and eclampsia contribute significantly to maternal deaths, forming part of the "deadly triad" along with hemorrhage and infection.**

HIGH-YIELD

- **Preeclampsia is the most dangerous hypertensive disorder in pregnancy due to multiorgan involvement and risk of progression to eclampsia.**

2. TERMINOLOGY AND CLASSIFICATION

Types of Hypertensive Disorders

- **Preeclampsia and eclampsia represent pregnancy-specific hypertensive syndromes with systemic involvement.**
- **Chronic hypertension is defined as hypertension present before pregnancy or before 20 weeks of gestation.**
- **Preeclampsia superimposed on chronic hypertension occurs when new features of preeclampsia develop in a hypertensive patient.**
- **Gestational hypertension is defined as new-onset hypertension after midpregnancy without proteinuria and resolves postpartum.**

IMPORTANT

- **Classification helps differentiate high-risk preeclampsia from less severe hypertensive states.**

3. DIAGNOSIS OF HYPERTENSION

Blood Pressure Criteria

- **Hypertension is diagnosed when systolic BP ≥140 mmHg or diastolic BP ≥90 mmHg.**
- **Korotkoff phase V is used to define diastolic blood pressure.**

Changes in Diagnostic Criteria

- **Previous criteria based on relative BP increase (e.g., +30 systolic or +15 diastolic) are no longer used.**
- **Blood pressure surveillance remains essential even when values are <140/90 mmHg due to risk of progression.**

Delta Hypertension (Page 3)

- **Delta hypertension refers to a sudden rise in mean arterial pressure within normal limits.**

- **Some women with delta hypertension may progress to preeclampsia, eclampsia, or HELLP syndrome.**

CLINICAL ALERT

- **Normal absolute BP values do not exclude preeclampsia if there is a significant rise from baseline.**

4. GESTATIONAL HYPERTENSION

Definition

- **Gestational hypertension is diagnosed when BP ≥140/90 mmHg develops after 20 weeks without proteinuria.**

Clinical Course

- **Nearly 50% of women with gestational hypertension later develop preeclampsia.**
- **Around 10% of eclampsia cases occur before detectable proteinuria appears.**

Outcome

- **If proteinuria does not develop and BP normalizes postpartum, it is termed transient hypertension.**

IMPORTANT

- **Absence of proteinuria does not exclude severe disease.**

5. PREECLAMPSIA SYNDROME

Definition

- **Preeclampsia is a pregnancy-specific syndrome affecting multiple organ systems due to endothelial dysfunction.**
- **Proteinuria is an important diagnostic marker reflecting systemic endothelial leak.**

Classification by Onset

- **Early-onset preeclampsia occurs before 34 weeks and is associated with severe placental dysfunction.**
- **Late-onset preeclampsia occurs at or after 34 weeks and is generally less severe.**
- **Preterm onset is defined as <37 weeks, while term onset is ≥37 weeks.**

Atypical Presentation

- **Some patients may lack proteinuria or fetal growth restriction but still have preeclampsia.**
- **Diagnosis may rely on other organ dysfunction markers.**

Diagnostic Criteria

Hypertension + Proteinuria

- **Proteinuria is defined as ≥300 mg/24 hours or urine protein/creatinine ratio ≥0.3.**
- **Dipstick ≥1+ may be used when no other method is available.**

OR Hypertension + End-Organ Dysfunction

- **Thrombocytopenia is defined as platelets <100,000/µL.**
- **Renal dysfunction is defined as serum creatinine >1.1 mg/dL or doubling of baseline.**
- **Liver involvement is indicated by elevated transaminases.**
- **Neurological symptoms include headache, visual disturbances, or seizures.**
- **Pulmonary edema indicates severe systemic involvement.**

HIGH-YIELD

- **Proteinuria is not mandatory if systemic organ dysfunction is present.**

6. SEVERITY CLASSIFICATION

Indicators of Severity

Severe Features

- **Systolic BP ≥160 mmHg or diastolic BP ≥110 mmHg indicates severe disease.**
- **Presence of headache, visual disturbances, or seizures indicates CNS involvement.**
- **Upper abdominal pain suggests hepatic involvement.**
- **Oliguria indicates renal compromise.**
- **Elevated creatinine and liver enzymes indicate organ dysfunction.**
- **Thrombocytopenia reflects platelet consumption.**
- **Pulmonary edema indicates fluid overload and capillary leak.**
- **Fetal growth restriction suggests placental insufficiency.**

Non-Severe Features

- **BP <160/110 mmHg without organ dysfunction indicates non-severe disease.**

IMPORTANT

- **Disease may progress rapidly from non-severe to severe, requiring close monitoring.**

Clinical Warning Signs (Page 6–7)

- **Headache and visual disturbances may precede eclampsia.**
- **Epigastric or right upper quadrant pain indicates hepatic ischemia or necrosis.**
- **Thrombocytopenia indicates worsening disease due to microangiopathic hemolysis.**

CLINICAL ALERT

- **Eclampsia can occur before, during, or after labor, including postpartum period.**

7. PREECLAMPSIA SUPERIMPOSED ON CHRONIC HYPERTENSION

Definition

- **Occurs when preeclampsia develops in a patient with preexisting hypertension.**

Diagnostic Criteria

- **New-onset proteinuria or worsening hypertension after 20–24 weeks suggests superimposed preeclampsia.**
- **Presence of end-organ dysfunction confirms diagnosis.**

Clinical Importance

- **Superimposed preeclampsia develops earlier and is more severe than primary preeclampsia.**
- **It is frequently associated with fetal growth restriction.**

8. INCIDENCE AND RISK FACTORS

Incidence

- **Preeclampsia occurs in 5–8% of pregnancies.**
- **Incidence is higher in nulliparous women compared to multiparous women.**

Demographic Factors

- **Younger nulliparous women have higher susceptibility.**
- **Older women are more likely to have chronic hypertension with superimposed disease.**

Racial and Genetic Factors

- **Incidence is higher in African American women and lower in some other ethnic groups.**
- **Genetic predisposition significantly influences disease risk.**

High-Risk Factors

- **Prior history of preeclampsia significantly increases recurrence risk.**
- **Chronic hypertension increases risk due to baseline vascular dysfunction.**
- **Diabetes mellitus contributes via endothelial damage.**
- **Obesity increases inflammatory and metabolic stress.**
- **Multifetal gestation increases placental mass and risk.**

Moderate Risk Factors

- **Nulliparity increases risk due to first exposure to placental antigens.**
- **Advanced maternal age (>35 years) moderately increases risk.**
- **Chronic kidney disease predisposes due to impaired renal function.**

Additional Factors

- **Antiphospholipid antibodies and SLE increase thrombotic risk.**
- **Assisted reproductive technologies are associated with increased incidence.**

HIGH-YIELD

- **Prior preeclampsia and chronic hypertension are the strongest predictors.**

9. ETIOPATHOGENESIS

Core Concept (Page 10–11)

- **The exact cause of preeclampsia remains unclear but involves abnormal placentation and systemic endothelial dysfunction.**

Predisposing Conditions

- **First exposure to chorionic villi increases risk due to immunological factors.**
- **Increased placental mass (e.g., twins, molar pregnancy) increases risk.**
- **Preexisting endothelial dysfunction predisposes to disease.**
- **Genetic predisposition influences susceptibility.**

Two-Stage Theory (Page 11)

Stage I: Placental Syndrome

- **Abnormal trophoblastic invasion leads to incomplete remodeling of spiral arteries.**
- **Spiral arteries remain narrow and high-resistance instead of becoming dilated.**
- **Reduced placental perfusion leads to hypoxia.**

Stage II: Maternal Syndrome

- **Placental ischemia triggers release of inflammatory and antiangiogenic factors.**
- **These factors cause systemic endothelial dysfunction.**
- **Endothelial damage leads to hypertension, proteinuria, and organ dysfunction.**

HIGH-YIELD

- **Placental ischemia → endothelial dysfunction → clinical preeclampsia.**

10. PLACENTAL PATHOPHYSIOLOGY

Normal vs Preeclampsia (Figure 40-3, Page 12–13)

Normal Pregnancy

- **Extravillous trophoblasts invade spiral arteries and replace endothelial and muscular layers.**
- **Arteries become dilated, low-resistance vessels allowing high blood flow.**

Preeclampsia

- **Trophoblastic invasion is incomplete.**
- **Spiral arteries remain narrow and high-resistance.**

- **Placental perfusion is reduced, causing hypoxia.**

CLINICAL MEANING

- **Reduced placental blood flow leads to fetal growth restriction and adverse outcomes.**

11. IMMUNOLOGICAL FACTORS

Mechanism (Page 13–14)

- **Loss of maternal immune tolerance to fetal antigens contributes to disease.**
- **Increased paternal antigen exposure (e.g., molar pregnancy) increases risk.**
- **Immune dysregulation leads to abnormal placentation.**

Cellular Immunity

- **Normal pregnancy shows Th2 dominance promoting tolerance.**
- **Preeclampsia shows increased Th1 activity causing inflammation.**

12. GENETIC FACTORS

Key Points (Page 14–15)

- **Preeclampsia is a multifactorial polygenic disorder.**
- **Risk is higher in daughters and sisters of affected women.**
- **Both maternal and fetal genes contribute to disease.**

IMPORTANT

- **No single gene is responsible; interaction of multiple genes determines phenotype.**

13. ENDOTHELIAL DYSFUNCTION

Mechanism (Page 15–16)

- **Placental ischemia releases cytokines and antiangiogenic factors.**
- **These cause systemic endothelial activation and injury.**
- **Endothelial dysfunction leads to:**
    - **Vasoconstriction**
    - **Increased vascular permeability**
    - **Coagulation activation**

Consequences

- **Capillary leak causes edema and proteinuria.**
- **Microvascular thrombosis causes thrombocytopenia.**
- **Reduced nitric oxide leads to vasoconstriction.**

14. VASOSPASM AND HYPERTENSION

Mechanism (Page 16)

- **Endothelial dysfunction leads to vasospasm.**
- **Vasospasm increases vascular resistance, causing hypertension.**
- **Reduced blood flow leads to ischemia and organ damage.**

Clinical Effects

- **Tissue ischemia causes necrosis and hemorrhage.**
- **Severe cases show reduced circulating blood volume.**

15. VASOACTIVE MEDIATORS

Prostaglandins (Page 16–17)

- **Reduced prostacyclin (vasodilator) and increased thromboxane (vasoconstrictor) cause imbalance.**
- **This imbalance increases vasoconstriction and platelet aggregation.**

Nitric Oxide

- **Nitric oxide is a vasodilator that maintains low vascular resistance.**
- **Reduced nitric oxide production contributes to hypertension.**

Endothelin

- **Endothelin is a potent vasoconstrictor elevated in preeclampsia.**
- **It contributes to vascular damage and renal injury.**

16. ANGIOGENIC IMBALANCE

Mechanism (Page 17)

- **Increased antiangiogenic factors reduce VEGF and PlGF activity.**
- **This leads to impaired vascular development and endothelial dysfunction.**

HIGH-YIELD

- **Angiogenic imbalance is a central mechanism in preeclampsia pathogenesis.**

QUICK REVISION

- **Preeclampsia is a multisystem disorder caused by endothelial dysfunction.**
- **Diagnosis requires hypertension with proteinuria or organ dysfunction.**
- **BP ≥140/90 mmHg defines hypertension in pregnancy.**
- **Severe disease is defined by BP ≥160/110 mmHg or organ damage.**

- **Proteinuria is not mandatory for diagnosis.**
- **Headache and visual symptoms are warning signs of eclampsia.**
- **Gestational hypertension can progress to preeclampsia.**
- **Superimposed preeclampsia occurs in chronic hypertension.**
- **Major risk factors include prior preeclampsia, obesity, diabetes, and hypertension.**
- **Pathogenesis follows a two-stage model: placental → maternal syndrome.**
- **Abnormal spiral artery remodeling leads to placental ischemia.**
- **Endothelial dysfunction causes vasospasm and organ damage.**
- **Prostaglandin imbalance favors vasoconstriction.**
- **Nitric oxide decreases and endothelin increases in preeclampsia.**
- **Angiogenic imbalance plays a central role in disease progression.**